

PRIYANUJ BORUAH

B.Tech Electronics Engineering (VLSI Design), Class of 2027

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SUMMARY

Final-year Electronics Engineering (VLSI Design) student who has taken multiple FPGA projects from specification and microarchitecture through synthesizable RTL, UVM verification with assertions and coverage, Quartus synthesis and static timing closure across all PVT corners, to bring-up on real silicon. GATE 2026 AIR 1626 (Instrumentation Engineering).

EXPERIENCE

Research Intern — CNN Inference Accelerator, *Centre for Nano Electronics & VLSI Design (CNVD), VIT Chennai* (May–Jul 2025)

Designed a fixed-point CNN inference core (convolution → quantization/ReLU → pooling) in Verilog for a Cyclone V FPGA; verified via self-checking Icarus Verilog tests (12/12, all pooling modes) and closed timing at 50 MHz (+4.97 ns slack, all PVT corners) in Quartus, 5% ALM utilization.

Industrial Maintenance Intern, *Indian Oil Corporation, Guwahati Refinery* (May–Jun 2026)

Investigated equipment downtime with structured root-cause methods across rotary and static refinery equipment; documented findings and SOPs for cross-functional engineering review. Researched on Pressure Sand Filters and their maintenance during high silt influx in the Brahmaputra during monsoons, in the Water Demineralization Plant.

Co-Founder & Team Manager, *Reva Solar Racing* (2023–2026)

Led a 20+ member team across parallel subsystem workstreams to competition deadlines; Best Team Spirit at SEVC Coimbatore.

SKILLS

RTL & Design: SystemVerilog, Verilog, synthesizable RTL, microarchitecture definition, FSMs, pipelining and retiming, arbitration, fixed-point arithmetic, RAM inference, clock-domain-crossing synchronizers, reset strategy

Verification: UVM (agents, sequencers, scoreboards, config objects), SystemVerilog Assertions, functional and code coverage

Protocols & Tools: AXI4, UART, QuestSim, Icarus Verilog, GTKWave, Quartus Prime, quartus_sta / TimeQuest, Platform Designer (Qsys), MATLAB, Cadence Tool Suite – NC Launch – Genus – Innovus, Sentaurus TCAD, VCS

Scripting & Environment: Python, Tcl, C/C++, Linux, Git/GitHub

PROJECTS

AXI4 Full Crossbar Interconnect — Cyclone V | *SystemVerilog, UVM, SVA, Questa, Quartus Prime, TimeQuest STA*

- Designed a **3×2 full AXI4 crossbar (AR/AW/W/B/R)** with QoS-weighted **round-robin arbitration**, ID-based out-of-order completion, up to 4 outstanding transactions per master, and **DECERR response for unmapped addresses**.
- Built the **UVM environment** from scratch: **master/slave agents, byte-exact race-checked scoreboard, covergroups over master × slave × burst × response × QoS, SVA protocol checkers**; 187/187 checks pass, with residual coverage gaps analysed per-coverpoint and structurally unreachable bins documented rather than hidden.
- Closed timing on all four PVT corners, **TNS = 0.000 across setup/hold/recovery/removal (139 MHz worst-case Fmax vs. a 50 MHz requirement, 165 ALMs)**; logged 7 root-caused bugs, including broadcast BREADY/RREADY colliding two slaves' handshakes under multiple outstanding transactions.

Systolic NPU with Reusable UVM Verification Environment | *SystemVerilog, UVM, Python, Questa, Quartus, Qsys, Nios II*

- Designed a parameterised INT8 weight-stationary systolic matrix-multiply accelerator (**PE array, weight-load FSM, skew/de-skew buffers, LOAD→COMPUTE→DRAIN→DONE control**) and took it from golden model to verified operation on silicon.
- Architected a **UVM environment** around a single config object driving both DUT parameters and testbench behaviour — **active/passive agents, virtual sequencer, scoreboard checked against one Python reference model**, functional coverage, and SVA including a cycle-exact ROWS+COLS–1 latency assertion; 3,276 checks, 0 failures.
- Integrated the unmodified core behind an Avalon-MM register wrapper driven by a Nios II soft core across a clock-crossing bridge; retimed 50 → 100 MHz at **3.109 ns setup slack, re-verified on hardware at 1,040/1,040 checks**. 25 bugs logged across simulation, synthesis and board bring-up.

Weibull-CFAR SAR Ship Detector — MATLAB Model to Cyclone V SoC (Capstone) | *Verilog, MATLAB, Python, Icarus Verilog, Quartus STA, Avalon-MM, ARM/Linux HPS*

- Implemented a streaming one-pixel-per-cycle detection pipeline (line buffer → incremental window sums → log-cumulant estimator → LUT → threshold compare) with backpressure handshaking, O(SLI × width) line memory instead of O(SLI²) registers, and LUT-folded constants removing all runtime division, log and exponential from the datapath.
- Wrote a **bit-accurate fixed-point MATLAB model** (0.0019% mean mismatch vs. floating point) as the direct RTL specification, generated golden vectors from it, and cross-checked the full chain against an independent Python model at 100.0000% exact match.
- Found via a dedicated STA run that the design had never met its 50 MHz clock (–34.968 ns worst setup slack, failing every corner) despite all functional tests passing; traced the paths and fixed two real RTL defects by **pipelining : reaching +3.050 ns slack, TNS = 0.000, ALMs down 52% → 49%, latency unchanged**. A full-chip rebuild then exposed a **–0.007 ns hold violation from an M10K/DSP placement accident**, closed via **placer seed** with zero RTL edits.

ACADEMICS & CERTIFICATIONS

B.Tech in Electronics Engineering (VLSI Design) | Vellore Institute of Technology, Chennai | 2023–2027 | CGPA: 8.04

- Introduction to Six Sigma** — SkillUp by Simplilearn.
- Introduction to Intellectual Property Categories and Compliance (Basics) and Addressing IP Challenges** — Purdue University Online.